H S Trinath Tammisetti

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.33
Nalanda Junior College	Class XII (Board of Intermediate Education AP)	2007	95.00%
St.Marys English Medium School	Class X (Board of Secondary Education AP)	2005	80.83%

SCHOLASTIC ACHIEVEMENTS

- Awarded Qualcomm fellowship-2024 as one of 15 recipients from a pool of 98 innovation proposals submitted across India.
- Institute Travel Grant Awarded to attend and present research at the ISCAS 2024 Conference.
- PhD Research Fellowship from MHRD, Govt. of India, 2020.
- Topped in M.Tech(IIT Patna)with highest CPI in Electrical Department.
- Qualified GATE thrice.
- Member of the INSTITUTION OF ELECTRONICS AND COMMUNICATION ENGINNERS (IETE).
- Received a meritorious scholarship amount of Rs.5000/- during my under graduation (B.Tech) from all over India .

POSITIONS OF RESPONSIBILITY

- Supported course instruction and laboratory sessions for Analog IC Design, Analog Electronics, and Analog Laboratory, including circuit-design guidance and experimental troubleshooting.
- Mentored undergraduate and graduate students on analog and RF research projects and supported technical reviews within the research group.

SKILLS

Analog / RFIC Design: VCDLs, PFDs, charge pumps, Nauta OTAs, active loop filters, N-path band-pass and band-stop filters, electrical-balance duplexers, and LNAs.

EDA & Simulation: Cadence Virtuoso for schematic capture and layout, Cadence Spectre/SpectreRF, and LTspice; experienced in DC, AC, transient, S-parameter, noise, noisefigure, linearity, PSS, PAC, and process-corner analyses.

Layout & Physical Verification: Full-custom transistor-level layout in Cadence Virtuoso; Calibre DRC/LVS; layout debugging and LVS issue resolution.

RF Measurement & Programming: VNA, spectrum analyzer, RF/baseband signal generators, oscilloscope; MATLAB and C; Ki Cad PCB design

RELEVANT COURSES

- RFIC Design
- Analog IC Design
- Introduction to VLSI Design

PROJECTS

Same-Channel Full-Duplex Receiver | 65 nm CMOS | Fabricated and Measured Designed a mixer-less receiver that extracts the desired signal from relative zero-crossing timing shifts between the transmitted and antenna signals,

eliminating a dedicated RF down conversion mixer and conventional multi-domain SI cancellers.

Design 1 - Voltage-Controlled Delay Line (VCDL): Designed a varactor-loaded inverter delay cell using an NMOS N-well varactor; achieved approximately 10 ps delay tuning over 500-700 mV and 55 ps/V gain near 600 mV.

Design 2 - TSPC Phase-Frequency Detector: Designed and transistor-sized a high-speed true-single-phase-clock PFD; verified operation from 10 MHz to 5 GHz with approximately 0.5 mW consumption across the 0.8-1.2 GHz target band.

Design 3 - Differential Charge Pump: Implemented a source-switched fully differential charge pump with low-voltage cascode biasing, reduced clock feedthrough, and matched 1 mA source/sink branches.

Design 4 - Active Loop Filter / OTA: Designed a six-inverter Nauta transconductor with 400fF integrating capacitors; achieved approximately 10 GHz unity-gain bandwidth while maintaining common-mode stability.

Design 5 - Full-Custom Layout and Physical Verification: Implemented the receiver layout in Cadence Virtuoso and independently completed Calibre DRC and LVS verification for the integrated analog/mixed-signal blocks.

Design 6 - Receiver Integration and Silicon Validation: Integrated input buffers, dual VCDLs, PFD, charge pump, and OTA loop filter in a 0.0125 mm2 core. The prototype consumes 26 mW from 1.2 V, operates from 0.8-1.2 GHz, supports 10 MHz signal bandwidth, and demonstrates 77 dB SI suppression for 0 dBm SI. Characterized port matching, delay-tovoltage transfer, closed-loop response, noise, gain, linearity, SFDR, and modulated-signal recovery using a VNA, spectrum analyzer, oscilloscope, RF source, and baseband signal generator.

Tunable FDD Duplexer Using Electrical Balance with N-Path Filters | IEEE ISCAS 2024 | 65 nm CMOS Schematic Design

Design 1 - N-Path Band-Pass Filter: Designed an 8-path passive BPF and selected switch and capacitor values for low-loss transmit-band operation; achieved less than 1 dB insertion loss and more than 13 dB rejection at 50MHz Tx-Rx frequency separation.

Design 2 - N-Path Band-Stop Filter: Designed a capacitor-shared 8-path BSF for receive-band reflection and close-in rejection; achieved more than 21 dB rejection with less than 2 dB insertion loss at 50 MHz Tx-Rx frequency separation.

Design 3 - MOS Switch Sizing: Sized low-threshold NMOS switches to W/L = 12 um/60 nm for approximately 12 ON resistance, balancing insertion loss and rejection in the N-path filters.

Design 4 - Electrical-Balance Hybrid: Designed and analyzed a capacitive-bridge electrical-balance coupler using ideal passive components, including 50 port matching and component selection for Tx/Rx signal combining and isolation.

Design 5 - RF Front-End Integration and Spectre Verification: Integrated the BPF/BSF cascades with the electricalbalance duplexer and verified tunability from 0.8-1.5 GHz, greater than 40 dB Tx-Rx isolation, less than 1.4 dB Tx insertion loss, and less than 3 dB Rx insertion loss using Cadence Spectre.

Variable-Gain CMOS Low-Noise Amplifier | Course Project

Designed and simulated a 2.4-2.5 GHz variable-gain CMOS LNA using Cadence Virtuoso; performed transistor sizing, DC biasing, impedance matching, and gain optimization, and evaluated S-parameters, noise figure, IIP3, bandwidth, and power trade-offs using SpectreRF.

PUBLICATIONS

H. S. Trinath Tammisetti, N. Malgundkar, and A. Kumar, "A Tunable FDD Duplexer using Electrical Balance with N-Path Filters," IEEE International Symposium on Circuits and Systems (ISCAS), 2024.

H. S. Trinath Tammisetti and A. Kumar, "A Same-Channel Full-Duplex Receiver Using Self-Interference Signal for Downconversion,"(Submitted to IoT Journal under Major Review).

H. S. Trinath Tammisetti and A. Kumar, "A Zero-Crossing-Based Time-Domain Receiver for Monostatic FMCW Radar," (Under preparation).

PATENT

Granted Patent: "A Frequency Division Duplexer (FDD) to Facilitate Wideband Isolation of Transmitting and Receiving Frequency," Indian Application No. 202441037093.

Published Patent: "Timing-domain receiver apparatus and method of extracting target-related information using the receiver apparatus," Indian Application No. 202641094471.

WORKING EXPERIENCE

MediaTek India Pvt. Ltd., Bengaluru | R&D Engineer / Senior Engineer | 2018-2020

Worked on wireless communication systems and 3GPP MAC/RLC specifications, contributing to protocol-level analysis, implementation support, and cross-functional engineering activities.